Max Hanrahan

62 Loveland St. Apt 9
Middletown, CT 06457

Phone: (814) 218-9634
Email: mhanrahan@wesleyan.edu

GitHub: max-hanrahan
LinkedIn: max-hanrahan

Education

Ph.D. Physics, Wesleyan University, 2022-2027 (expected).

B.A. Physics, Hamilton College, 2022.

Employment

Wesleyan University, Graduate Research Assistant, 2022–Present. Advisor Dr. Francis W. Starr. Develop and execute molecular dynamics simulations using LAMMPS, C++, and Python to investigate the structure, dynamics, and phase behavior of complex polymer systems. Projects include modeling star-shaped polymers in linear polymer melts and analyzing the role of dynamic, short-lived bonds in compatibilizing immiscible symmetric polymer mixtures and altering surface tension. This work has resulted in publications in *Macromolecules* and *The Journal of Chemical Physics*.

National Renewable Energy Laboratory, REU, 2022. Advisor Dr. Christopher Hoehne. Quantified the effectiveness of an e-bike pilot program in Denver, CO, using the novel Mobility Energy Productivity (MEP) metric and live trip data from an open-source mobile application (OpenPATH). Analyzed open source travel data using Pandas in Python. Contributed data analysis and visualization code to the OpenPATH code repository. Published findings in the Transportation Record Review (TRR) Journal, and presented a poster at the Transportation Research Board (TRB) annual meeting 2023.

Undergraduate Researcher, 2022. Advisor Professor M Smith. Investigating a magnetohydrodynamic, fully general-relativistic (MHDGR) black hole simulation by examining the evolution of plasma in its accretion disk. The model is time-dependent, fully 3d global, and tracks volume density and magnetic field strength. Specifically, we are interested in investigating the hypothesis that magnetic Rayleigh-Taylor instability plays a significant role in angular momentum transport in the thin, magnetically arrested disk (MAD) of a spinning black hole.

National Institute of Standards and Technology, REU, 2021. Advisor M. Gaitan. Our team developed and tested a method to calibrate a 3-axis accelerometer and gyroscope under dynamic conditions, using pendulum excitation. I accounted for offset by measuring the accelerational intrinsic sensitivity along each of the three accelerometers. Additionally, I designed and built a pendulum in my home to excite the accelerometers and gyroscopes to calibrate the rotational sensitivity of the gyroscope under dynamic conditions. I then performed a calibration analysis by fitting the measured acceleration and rotation data to the equation for pendulum motion. Uncertainty was determined from the covariance of the parameters in the fit.

Undergraduate Researcher, 2020 Advisor Professor M Smith. Worked to improve the MADNT black hole simulation code used in the research of Avara et al. and Marshall et al. Updated analysis code from Python2 (no longer supported) to Python 3 (supported). Using yt-project and PyVista, we successfully produced a 3D rendering script to visualize the mass density of the magnetically arrested disk surrounding the spinning black hole, as well as the emergent magnetic fields.

Publications

Hanrahan, M. K., Grandfield, E., Kumar, S. K., Douglas, J. F., & Starr, F. W. (2026). *Phase behavior, self-assembly, and interfacial tension of a dynamically linked polymer blend*. **The Journal of Chemical Physics**, 164(19), 194907.
Honors: Published in the Special Collection: Festschrift in Honor of Kurt Kremer. Selected as an **Editor's Pick** and featured in an **AIP Scilight**.

Castro Rubio, C. A., Fan, J., **Hanrahan, M. K.**, Douglas, J. F., & Starr, F. W. (2023). *Structure and Dynamics of Composites of Star Polymers in a Linear Polymer Matrix*. **Macromolecules**, 56(23).

Hoehne, C., **Hanrahan, M.**, & Shankari, K. (2023). *Mobility Energy Productivity and Equity: E-Bike Impacts for Low-Income Essential Workers in Denver*. **Transportation Research Record: Journal of the Transportation Research Board**.

Invited Talks, Conference, and Workshop Proceedings

Polymer Physics Gordon Research Conference, *South Hadley, MA*. Presented poster entitled *Phase Behavior, Self-Assembly, and Interfacial Tension of a Dynamically Linked Polymer Blend*, Summer 2026.

Regional Meeting of APS Soft Matter, *Boston, MA*. Presented talk entitled *Surface Tension and Phase Behavior in a Dynamically-Linked Polymer Network*, Fall 2024.

Annual Meeting of the American Physical Society, *Minneapolis, MN*. Presented talk entitled *The Effect of Chain Stiffness on the Dynamics of Polymer/Star Polymer Composites*, Spring 2024.

Summer School on Soft Solids and Complex Fluids, *University of Massachusetts Amherst* Summer 2023. Annual summer school and poster-session for graduate students in soft matter and statistical physics hosted by the Physics Department at UMass Amherst, Summer 2023.

Transportation Research Board Annual Meeting 2023 *Washington, D.C.* Presented poster of research findings entitled *MEP and Equity: E-bike Impacts on the MEP Scores of Low-income Essential Workers in Denver*, Winter 2023.

Hamilton College Senior Thesis Defense, *Analyzing Angular Momentum in Spinning Black Holes Undergoing Magnetic Rayleigh-Taylor Instability*, Spring 2022.

Plenary Session *National Institute of Standards and Technology (NIST), Gaithersburg, MD*, Fall 2020. Presented findings on research I conducted during my REU at NIST Gaithersburg to fellow student and professional researchers in a talk entitled *Gyroscope Calibration via Pendulum-Based Excitation*.

Honors and Activities

President, Wesleyan University Graduate Student Association, 2026-2027, Chaired executive governance and served as the primary administrative liaison for the Graduate Student Association, managing council operations, representative appointments, dispute resolution, and major academic events like the Graduate Student Speaker Series.

Treasurer and Acting Vice President, Wesleyan University Graduate Student Association, 2025-2026, Oversaw financial operations and budget accounting for the Graduate Student Association, serving as fiscal advisor to the Executive Board on funding requests, reimbursement approvals, and constitutional governance procedures.

Top 100 Finalist, Summer of Math Exposition (SoME1) 2021 Created an explanatory mathematics video, based on a lecture from Hamilton College Professor Seth Major. I programmed animations using Python-based open source animation software Manim and was recognized by a peer-review process as one of top 100 of over 13,000 entries.

The Charles A. Dana Prize Scholarship, 2019-2022, awarded annually to approximately ten sophomores at the end of their first year in recognition of academic achievement, character, and leadership. The recipients of this award who have financial aid will have the scholarship part renewed yearly, until senior year, as long as they continue to fulfill the requirements.

Teaching Experience & Outreach

Instructor, *National Education Opportunity Network* Taught Intro to Python Programming weekly to approximately 20-25 high school students for 45 minutes via Zoom.

Research Project Leader, *Warrior Scholar Project* Guided 5 former veterans on a coding research project using NumPy and Pandas in Python. Held class 8:45-10:45am from Monday to Friday, using a mix of prepared lectures and active learning sessions.

Teaching Assistant, *Wesleyan University*. Instructed laboratory experiments and provided individualized academic support through targeted feedback on assignments and dedicated weekly office hours.

Physics Grader, *Hamilton College*. Graded weekly student problem sets and offered constructive feedback. As head grader I addressed student comprehension issues with faculty supervisor.

Laboratory Learning Assistant, *Hamilton College*. Served as a learning/teaching assistant in the Physics 100 and Physics 200 labs. Facilitated learning by asking questions about ongoing student experiments. Graded student work and offered constructive feedback.

Technical Skills

Programming & Scripting: Python, C++, R, LaTeX

Simulation & Modeling: LAMMPS, Quantum ESPRESSO, Pacemaker (ACE), Etomica

Data Analysis & Visualization: Pandas, NumPy, OVITO, PyVista, yt-project

Software & Tools: Git, GitHub